Overcurrent in Discharge (OCD) and Short Circuit in Discharge (SCD) are implemented using sampled analog comparators that run at 32 kHz, and that continuously monitor the voltage across (SRP – SRN) while the device is in NORMAL mode. Upon detection of a voltage that exceeds the programmed OCD or SCD threshold, a counter begins to count up to a programmed delay setting. If the counter reaches its target value, the SYS_STAT register is updated to indicate the fault condition, the FET state(s) are updated as shown in 表 8-1, and the ALERT pin is driven high to interrupt the host.

The protection fault threshold and delay settings for OCD and SCD protections are configured through the PROTECT1 and PROTECT2 registers. See *节 8.5* for details about supported values.

Overvoltage (OV) and undervoltage (UV) protections are handled digitally, by comparing the cell voltage readings against the 8-bit programmed thresholds in the OV and UV registers.

The OV threshold is stored in the OV_TRIP register and is a direct mapping of 8 bits of the 14-bit ADC reading, with the upper 2 MSB preset to “10” and the lower 4 LSB preset to “1000”. In other words, the corresponding OV trip level is mapped to “10-XXXX-XXXX – 1000”. The programmable range of OV thresholds is approximately 3.15 to 4.7 V, but this is subject to variation due to the (GAIN, OFFSET) linear equation used to map the ADC values.

The UV threshold is stored in the UV_TRIP register and is a direct mapping of 8 bits of the 14-bit ADC reading, with the upper 2 MSB preset to “01” and lower 4 LSB preset to “0000”. In other words, the corresponding UV trip level is mapped to “01-XXXX-XXXX – 0000”. The programmable range of UV thresholds is approximately 1.58 to 3.1 V, but this is subject to variation due to the (GAIN, OFFSET) linear equation used to map the ADC values.

Protection	Upper 2 MSB	Middle 8 Bits	Lower 4 LSB
OV	10	Set in OV_TRIP Register	1000
UV	01	Set in UV_TRIP Register	0000

备注

To support flexible cell configurations within BQ76920, BQ76930, and BQ76940, UV is ignored on any cells that have a reading under $UV_{MINQUAL}$. This allows cell pins to be shorted in implementations where not all cells are needed (for example, 6-series cells using the BQ76930).

Default protection thresholds and delays are shown in the register description at the end of this data sheet. These are loaded into the digital register (RAM) of the device when the device enters NORMAL mode. These RAM values may then be overwritten by the host controller to any other values, which they will retain until a POR event. It is recommended that the host controller reload these values during its standard power-up and/or reinitialization sequence.

To calculate the correct OV_TRIP and UV_TRIP register values for a device, use the following procedure:

1. Determine desired OV.
2. Read out *[ADCGAIN]* and *[ADCOFFSET]* from their corresponding registers. Note that ADCGAIN is stored in units of µV/LSB, while ADCOFFSET is stored in mV.
3. Calculate the full 14-bit ADC value needed to meet the desired OV and UV trip thresholds as follows:
   a. OV_TRIP_FULL = (OV – ADCOFFSET) ÷ ADCGAIN
   b. UV_TRIP_FULL = (UV – ADCOFFSET) ÷ ADCGAIN
4. Remove the upper 2 MSB and lower 4 LSB from the full 14-bit value, retaining only the remaining middle 8 bits. This can be done by shifting the OV_TRIP_FULL and UV_TRIP_FULL binary values 4 bits to the right and removing the upper 2 MSB.
5. Write OV_TRIP and UV_TRIP to their corresponding registers.

Both OV and UV protections require the ADC to be enabled. Ensure that the *[ADC_EN]* bit is set to 1 if OV and UV protections are needed.